

Activity 1.2.1 A-door-able Hens

PLTW

A-DOOR-ABLE HENS



Free Birds



Ins and Outs



Conclusion

Free Birds

GOALS	MATERIALS	RESOURCES
• Differentiate between the four types of motion. • Identify input and output motion of everyday household devices. • Design a simple device that takes advantage of motion types.		
GOALS	MATERIALS	RESOURCES
• VEX® V5 POE Kit		

GOALS	MATERIALS	RESOURCES
• VEX® Parts • VEX® V5 Build Support Guides • Activity 1.2.2 A-door-able Hens (Downloadable PDF)		

Fran's flock of egg-laying hens is producing record numbers of eggs! She is reaping the rewards of her hard work and dedication, while ensuring her hens have everything they need to be happy, healthy, and productive layers.

However, time takes a toll on farm infrastructure. Water hoses crack and need replacement. Fences need constant mending. Fruit tree trellises require frequent adjustment and tightening. Most recently, the latch to the chicken coop door failed. Now the chickens come and go as they please, making them vulnerable to predator attacks.



Fran reached out to the engineering community to see whether they can help design a better latch for the door to keep the “free birds” in. After you learn about some basic types of motion, you will be well prepared to help solve this design problem.

Types of Motion

To design an effective latch, you need to understand the different types of motion. Engineers typically consider four types of motion: reciprocating motion, linear motion, oscillating motion, and rotational motion.

Type of Motion	Definition
reciprocating motion	A repeated back and forth motion along a straight line.

linear motion	One-dimensional motion along a straight line.
oscillating motion	A repeated, curved, back and forth movement that pivots about an axis.
rotational motion	Motion about a point or axle.

1

Review the Four Types of Motion information.

Four Types of Motion

Learn more about...

RECIPROCATING

LINEAR

OSCILLATING

ROTATIONAL

A repeated, back-and-forth motion along a straight line. In a motorcycle, the pistons in the engine reciprocate (repeatedly move up and down) while the engine is running.



RECIPROCATING

LINEAR

OSCILLATING

ROTATIONAL

A straight line motion, in one direction such as a car driving down a road.



RECIPROCATING

LINEAR

OSCILLATING

ROTATIONAL

A repeated, curved, back-and-forth movement that pivots about an axis. A swing goes back and forth along an arc. The radius of the arc is the distance between the bottom of the swing and the fastener at the top that connects the chains of the swing to the horizontal beam.



RECIPROCATING

LINEAR

OSCILLATING

ROTATIONAL

A continuous motion around a central axis, such as how a potter's wheel rotates continuously in one direction.



2

With a partner, brainstorm a list of examples of each of these types of motion.

3

Compare your list with that of another pair of students.

Ins and Outs

Many common tools and devices have one type of input motion and one type of output motion. A bicycle (Figure 1) is a perfect example.

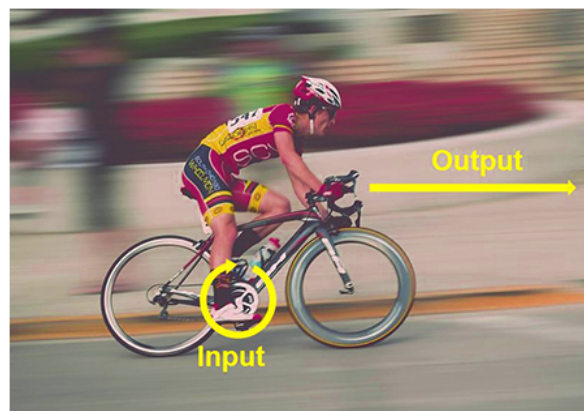


Figure 1. Cyclist's input (rotational motion) and output (linear motion)

The cyclist rotates the pedals around a central axis (in this case, called a bottom bracket) of the bicycle. This action is a rotational motion input. The final output is that the bicycle is propelled forward in a straight line or linear motion.

4

With your partner, brainstorm a list of at least five other simple devices. For each one, discuss and describe the input and output motions.

Need a Hint? —

Think of common handheld power tools or kitchen appliances.

Latch Onto a Design!

Now that you know a little bit about the types of motion, it is time to design and build a simple latch mechanism to help keep the chickens safe.

5

With a partner, brainstorm a list of common latch designs.

- Search the web for inspiration if needed.
- Generate simple sketches of these latch designs.
- Indicate the types of input and output motions for each latch, understanding that they might be the same.



6

Choose one design that you would like to make using VEX® parts. Construct that latch design. Use [**VEX®**](#) [**Parts**](#) and [**VEX® V5 Build Support Guides**](#) as necessary.

7

Exchange your latch with another group. Describe the input and output motions of each others' latches.

8

Design an entirely new latch. Try to avoid using commercially available latch designs, and instead focus

on something entirely new. In this case, the latch does not need to be the most efficient design possible.

- Generate simple sketches.
- Indicate the types of input and output motion.

9

Use VEX parts to construct your latch design.

Conclusion

Question 1

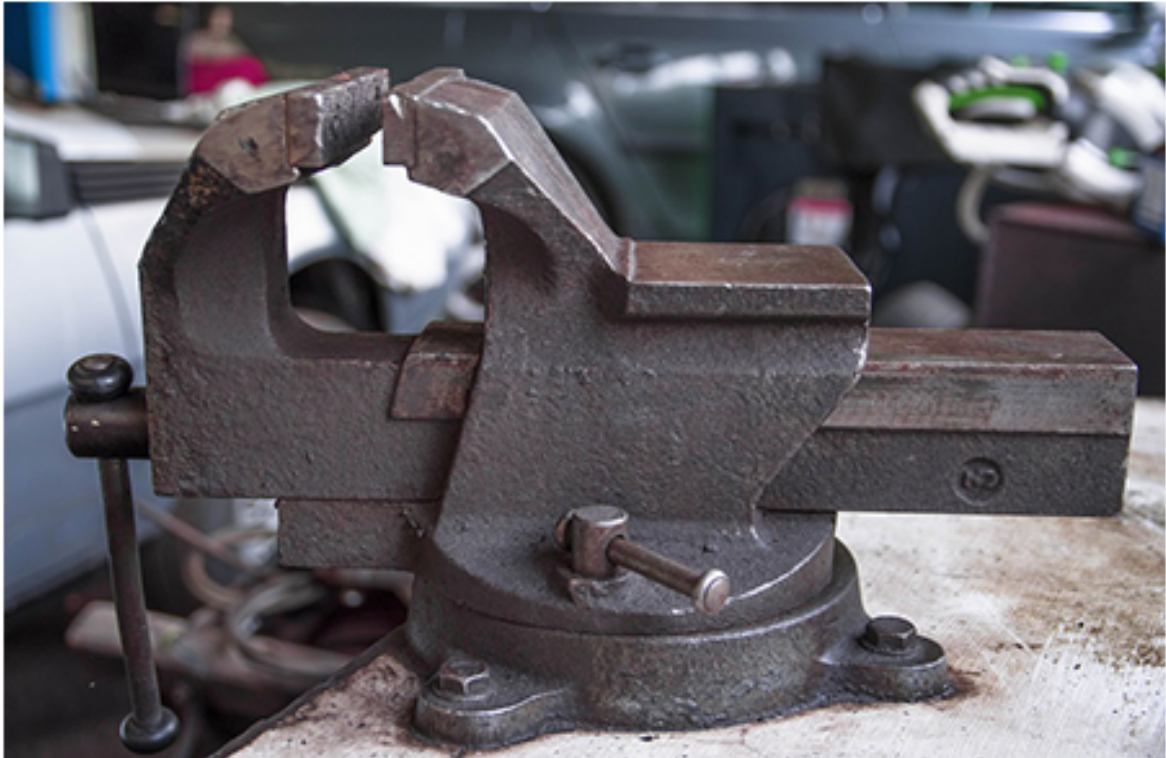
Describe how oscillating and rotational motions are both similar and different. Do the same for linear and reciprocating motions.

Question 2

Describe the input and output motions of a jigsaw, a vise, and a hi-hat cymbal. Provide justification for your choices.



Jigsaw



Vise



Hi-hat Cymbal